Is There an Optimal Movie Length for High Ratings?

In this project, I aim to explore whether movie length has an effect on audience ratings and whether there is an optimal duration that leads to higher ratings. While longer movies may allow for more detailed storytelling, they may also risk losing the viewer's attention. Similarly, shorter movies might be more engaging but may lack depth. Because of this, I am interested in understanding if there is a certain range of movie length that performs best in terms of ratings.

For this project, I will use publicly available movie datasets, such as those from Kaggle (for example IMDb or TMDb datasets). These datasets include information such as movie duration, ratings, genres, and other related features.

To enrich the analysis, an additional dataset will be incorporated. This may include variables such as movie budget, revenue, or release year. These additional features will help provide deeper insights into the factors that may influence movie ratings.

The dataset is expected to include thousands of movies (for example, more than 5,000 entries), which will allow for meaningful statistical analysis. The main variables I will focus on are movie duration and rating, but I may also consider additional variables such as genre or number of votes to better understand the results.

In the analysis part, I will first perform exploratory data analysis (EDA) to examine the relationship between movie length and ratings. I will use visualizations to observe patterns and trends. Then, I will apply regression methods to see whether there is a relationship between these variables and to investigate if an optimal movie length exists.